Gregory Lapit

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EDUCATION

Harvard University Cambridge, MA
A.B. in Physics, with Secondary in Computer Science *June 2024*
University of Alabama Tuscaloosa, AL
M.A. in Communications *June 2026*

WORK EXPERIENCE & TECHNICAL PROJECTS

Advanced Computational Materials Engineering Lab Tuscaloosa, AL
Graduate Simulation Researcher *September 2024 - August 2026*

- Refactored core simulation package for polycrystal plasticity, resolving quadratic scaling bottlenecks in numerical step routines to reduce average execution runtimes by 45% (110 mins → 60 mins).
- Engineered high-throughput data validation and post-processing pipelines; optimized parallel file system I/O using hierarchical data formats (HDF5), slashing output payload storage by 50% (9 GB → 4.5 GB).
- Automated distributed execution of 840+ cluster simulations via Slurm, writing custom Python and Bash orchestration tooling that replaced days of manual job configuration with zero-touch execution.
- Built automated QA and integration testing suites to validate simulation integrity across concurrent multithreaded jobs, ensuring data consistency across distributed compute nodes.
- Authored peer-reviewed computational journal paper (under review) detailing user accessibility, solver performance, and scalable architecture for complex differential deformation modeling.

Solitude – End-to-End Game Systems Development Tuscaloosa, AL
Solo Game Developer *August 2025– August 2026*

- Designed, built, and shipped a modular, full-stack game architecture in 12 months, managing end-to-end product lifecycle from technical scoping through continuous playtesting and deployment.
- Built 5+ decoupled, event-driven gameplay engines (combat state machines, dynamic inventory, branching dialogue, and persistent progression saves), prioritizing maintainability and low latency.
- Implemented user-driven feedback loops and telemetry, conducting iterative sprints based on user testing data; Maintained production hygiene using Git and CI/CD workflows

Boeing – Maintenance Ground Operations Systems Team Seal Beach, CA
Aerospace Support Equipment Design Engineer Intern *June 2023 - September 2023*

- Collaborated with maintenance and ground-crew engineers to analyze field constraints and model Boeing 737 support systems in CATIA; verified and integrated technical assets into enterprise asset databases.
- Designed and prototyped field-deployable maintenance tooling for landing gear fastener assemblies, directly resolving operational turnaround bottlenecks.
- Authored technical engineering change notices for patented aircraft ground-support tools.

Hoffman Laboratory Cambridge, MA
Undergraduate Thin Film Researcher *June 2022 - May 2024*

- Owned end-to-end delivery of a novel precursor inlet deposition system, solving an industry-wide synthesis bottleneck of uncontrolled sublimation in ultra-high vacuum (UHV) systems.
- Designed, fabricated, and integrated custom inlet hardware compatible with legacy molecular beam epitaxy (MBE) chambers, enabling stable, controlled introduction of volatile metal-organic precursors.
- Instrumented real-time closed-loop controls (PID regulation, ion gauges, linear manipulators), iteratively tuning deposition parameters to increase growth while maintaining crystal quality.
- Validated film structural integrity via nanoscale spectroscopy (AFM, SEM, XPS); delivered comprehensive technical documentation and presented project outcomes to 300+ researchers at the American Physical Society (APS) Conference.

ADDITIONAL ACTIVITIES

NCAA Division I Track & Field Student-Athlete Tuscaloosa, AL
University of Alabama and Harvard Long Sprinter *August 2019 – June 2026*

- Harvard University Record Holder in 2 Events | 5x Ivy League Champion | NCAA Division I National
- Devoted 20+ hours/week to elite athletic preparation and team leadership while maintaining a rigorous computational load.

SKILLS: Python, C++, React, TypeScript, Git, GLSL, Catia, Solidworks, SQL, Unity, Godot